Reprocessing the Past

Shifting Search and Rescue from Probability to Deterministic Physics.

SYROTUCK SAR DATA METRIC

- Of subjects found dead, 74% expire within 48 hours.
- Subjective search grid POD (Probability of Detection) peaks at 80% and fails entirely against environmental masking.

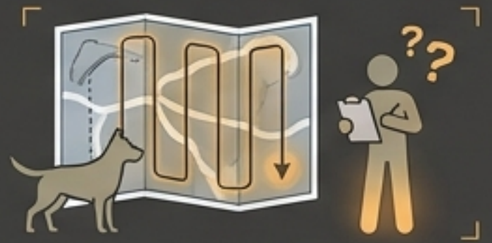
Forensic Finders:
Forensic Finders:Historical Forensic Material Reanalysis (HFMR) Engine

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SUBJECTIVE SEARCH VS. DETERMINISTIC PHYSICS

SUBJECTIVE PROBABILITY

Methodology:
Visual grid sweeps, K9 scent tracking, and subjective Probability of Detection (POD) estimates.



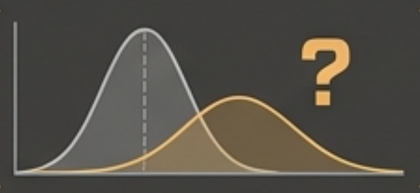
Treatment of Environment:
Defeated by terrain masking (snow bridges, dense rootballs, and subterranean vaults).



Evidence Status:
Highly subject to human bias, memory degradation, and perishable scent trails.

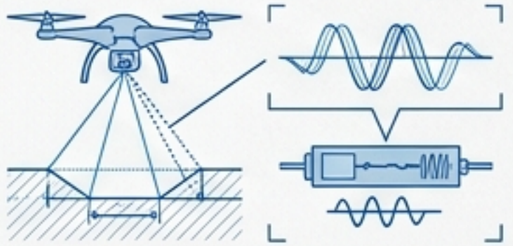


Outcome:
Guesses based on behavioral probability.

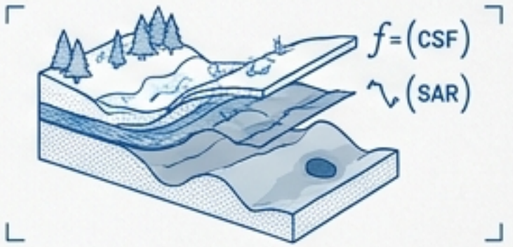


DETERMINISTIC PHYSICS

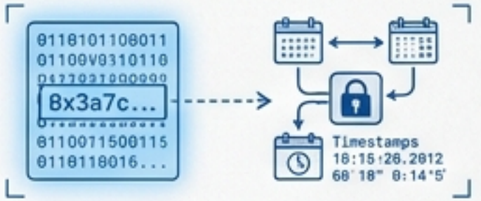
Methodology:
Multi-modal physics inversion, LiDAR bare-earth differencing, and dual-comb laser metrology.



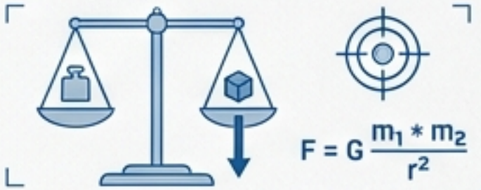
Treatment of Environment:
Mathematically strips masking variables (CSF filtering, synthetic aperture radar phase reconstruction).



Evidence Status:
Immutable, cryptographically hashed (SHA-256) data ledgers proving exact spatial timelines.



Outcome:
Calculated outcomes based strictly on mass, composition, and gravity.



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The Deterministic Funnel Architecture

Layer 1: Macro (Topography)

LiDAR & SAR Inversion

Cloth Simulation Filters (CSF) mathematically strip away canopy and rootbound blind spots to reveal bare-earth anomalies and clandestine ditch wells.

Layer 2: Meso (Spectral Illumination)

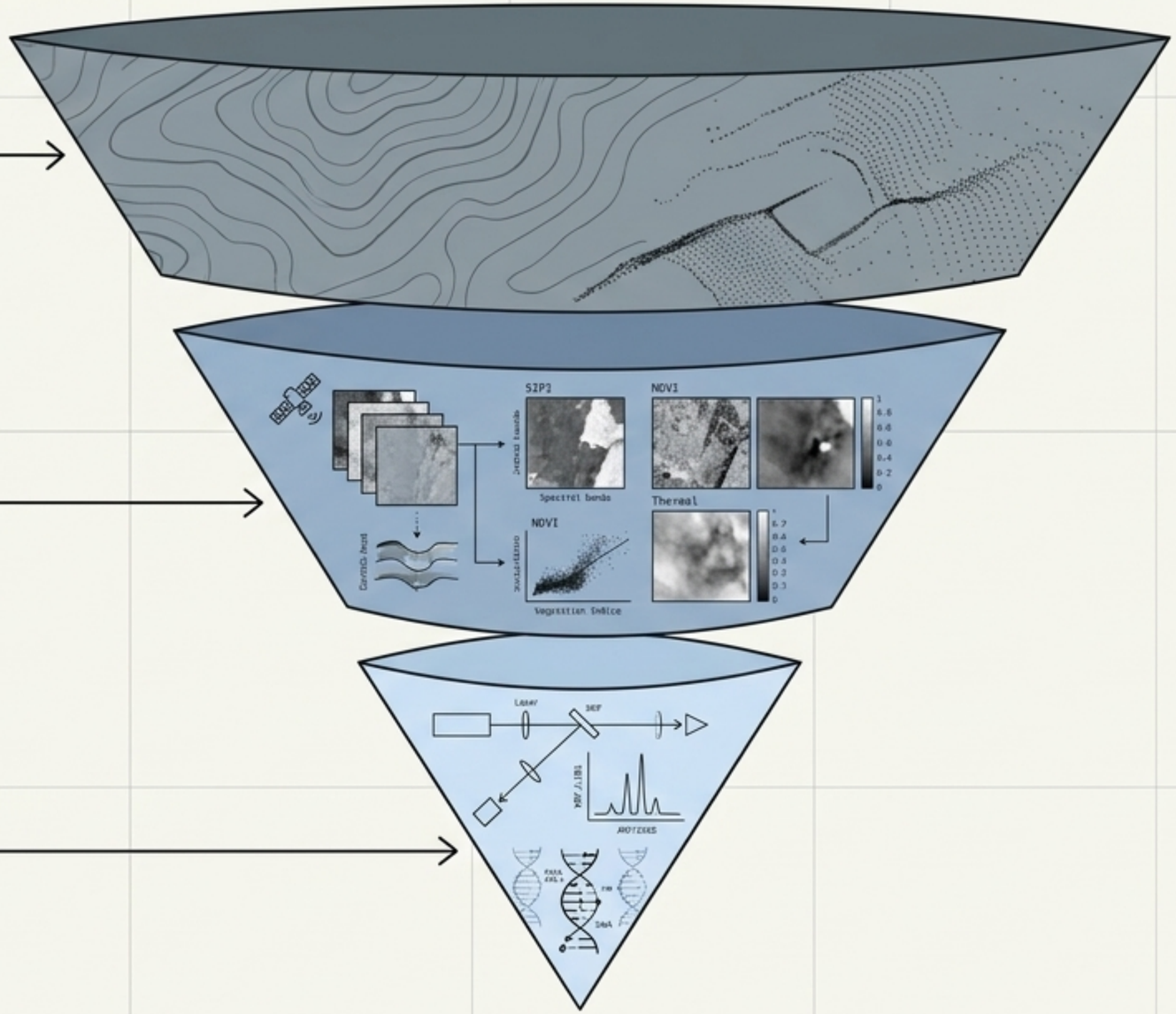
Year-over-Year Macro Illumination

Multispectral satellite data detects localized nitrogen flushes and chlorophyll suppression (SIPI/NDVI) acting as remote proxies for organic decomposition.

Layer 3: Micro (Material Identification)

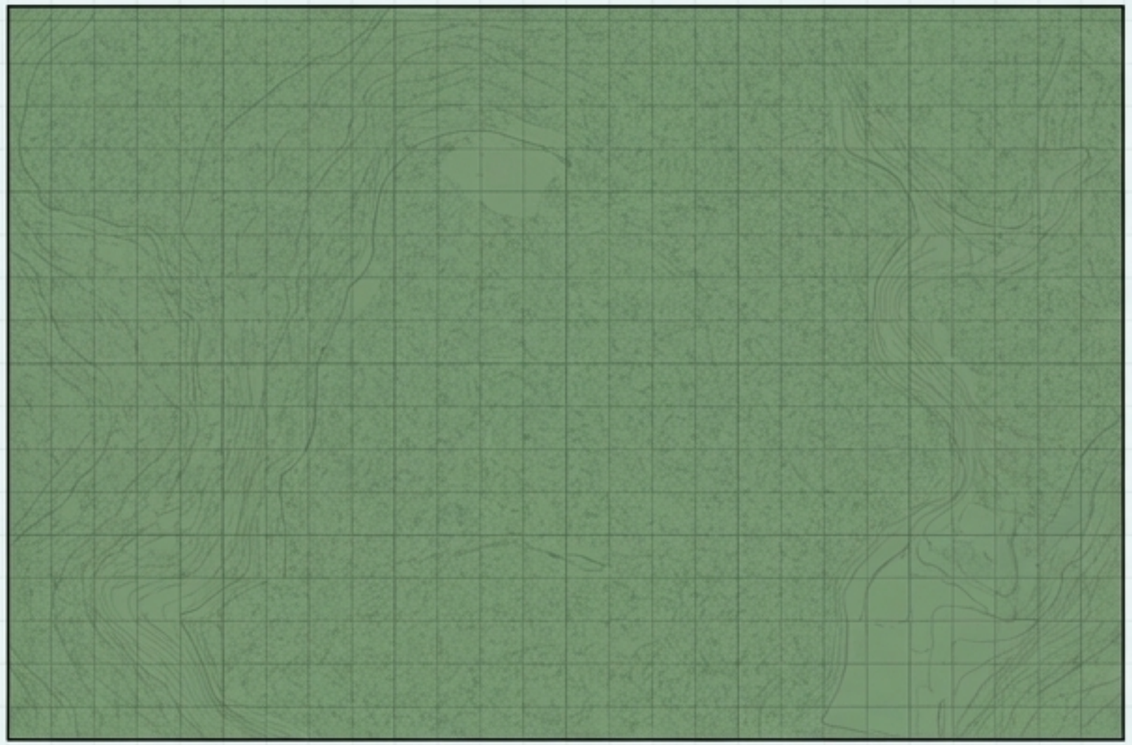
Atomic-Level & Bioinformatics

Dual-comb laser interferometry (4.6 µm ranging), XRF/LIBS spectroscopy for precise metal composition, and Massively Parallel Sequencing (MPS) for DNA alignment.

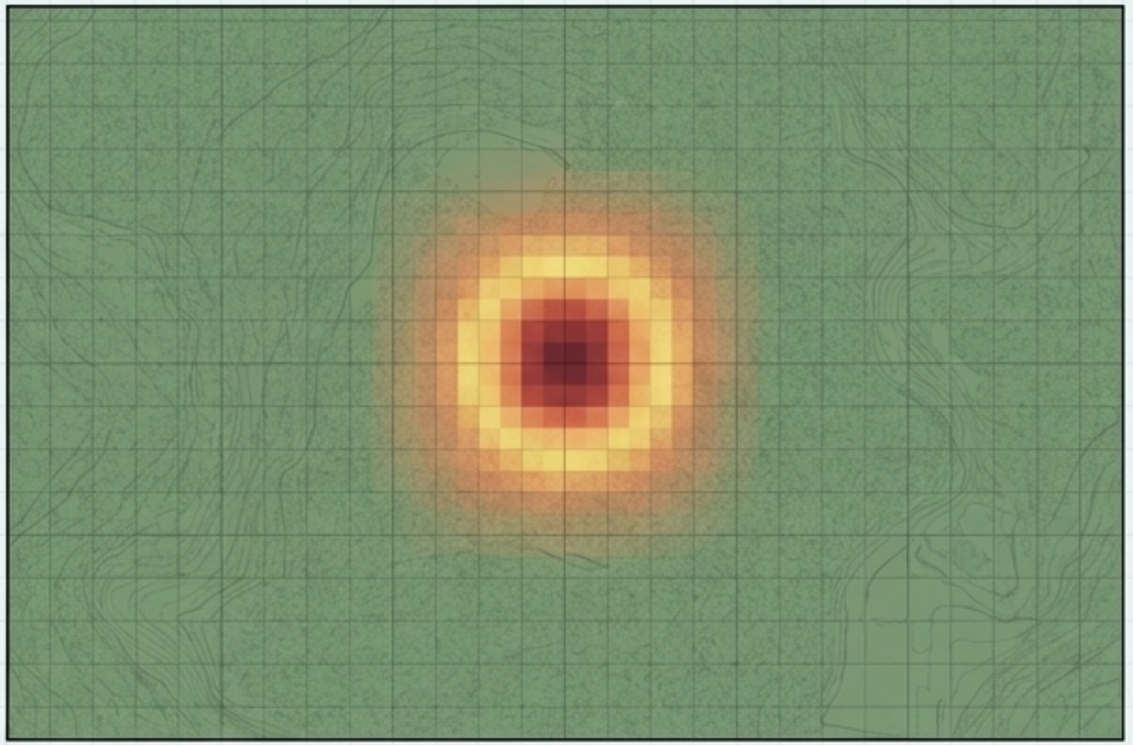


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Meso-Scale Atomic Detection via Macro Illumination



Year 1 Baseline: Standard Baseline Chemistry



Year 3 Bloom: Localized Nitrogen Flush

Human decomposition releases massive, highly concentrated atomic nitrogen into the soil. We write deterministic raster math scripts to compute vegetative indices (Normalized Difference Vegetation Index / Structure Insensitive Pigment Index). A sudden, highly localized spike illuminates the exact coordinates of atomic-level cellular breakdown, completely bypassing the need for physical searchers.

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Multi-Modal Physics Inversion Engine

Sensor Technology	Physical Measurement	Forensic Output
Synthetic Aperture Radar (SAR)	**Measures:** Microwave backscatter/phase	**Output:** Detects conductive/reflective anomalies through obscuration. Cannot identify exact elemental composition alone.
Dual-Comb Laser Interferometry	**Measures:** Optical phase/frequency	**Output:** Extremely precise exposed-surface geometry (nanometer-scale precision).
XRF / LIBS Spectroscopy	**Measures:** Element-specific emission	**Output:** Deterministic identification of exact metal composition (brass, aluminum, copper).
Hyperspectral Imaging	**Measures:** Nonlinear optical response	**Output:** Detects chemical vegetation shifts and localized nitrogen blooms.

THE FUSION MATRIX

No single sensor solves a case. The HFMR algorithm generates a weighted residual matrix—calculating the exact probability that an anomaly is metallic, independent of the probability that it contains human remains.

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The Farber Anomaly: TBI and Terminal Disorientation

1 The Disappearance (2015, Tamaqua, PA)

Jesse Farber reports being chased by “10 to 11 coyotes.” Panic suggests perceived threat.

2 The Coroner’s Factor

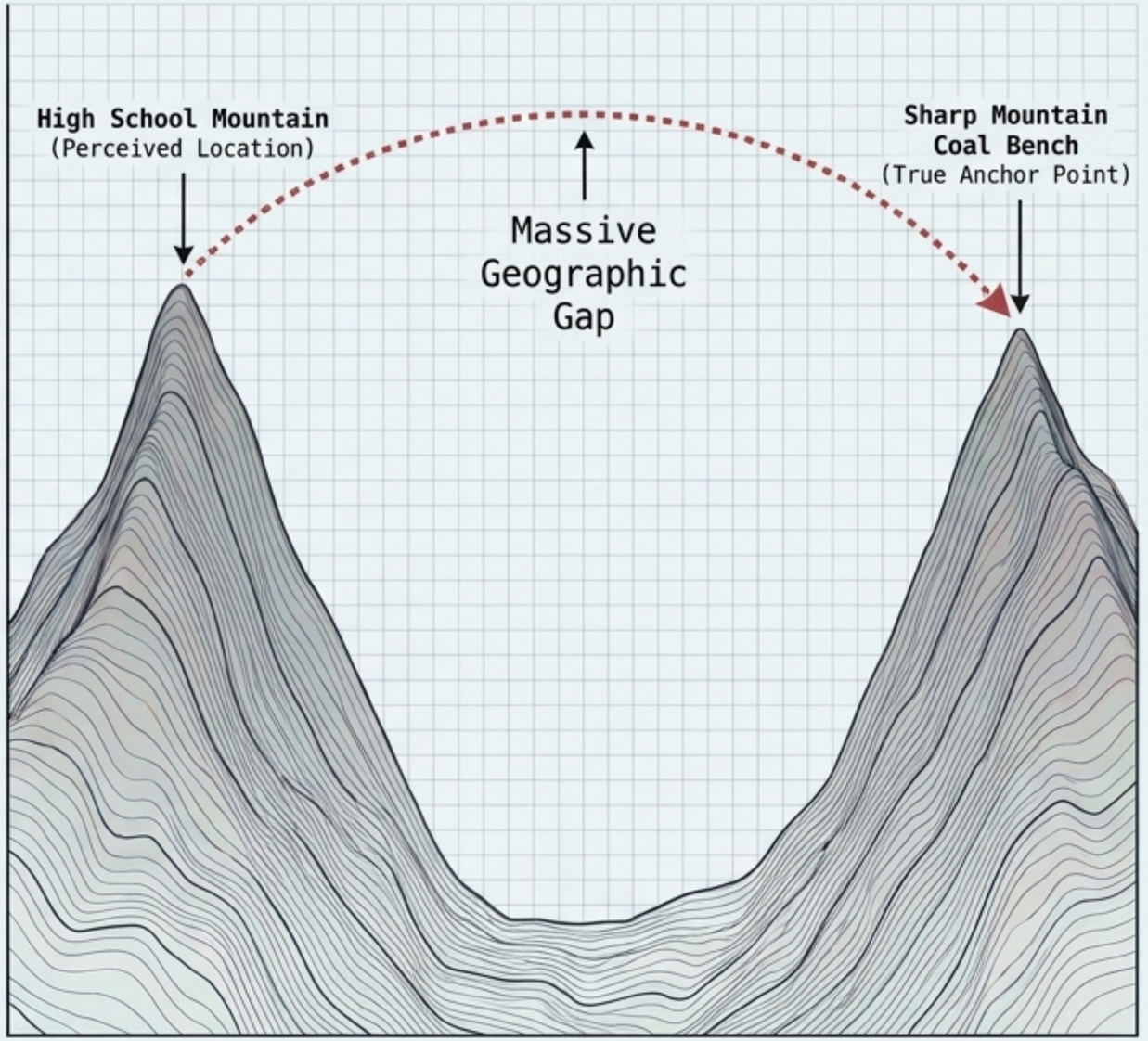
Traumatic Brain Injury (TBI) confirmed.

3 The Spatial Disconnect

TBI drove extreme paranoia and terminal disorientation, causing Farber to physically mix up the high school mountain and the elementary school mountain.

4 The Deposition Point

Driven by disorientation, the subject scaled a small, skinny tree on the edge of a rugged, private coal bench on Sharp Mountain.



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Deterministic Separation: Mass + Composition

Tree Canopy Deposition Point

Farber climbs the tree. Disorientation leads to haphazard anchoring (rock wedged in sweatpants leg as a weight/weapon).

PATH A: SYNTHETIC ARTIFACTS

Composition:

Nylon (Backpack), Cotton/Polyester (Sweatpants), Metal/Glass (Phone).



Physics:

Low mass, high snag potential.

Outcome:

Artifacts remain suspended in the tight branches of the skinny tree, enduring wind shear and degrading slowly over 11 years.

PATH B: ORGANIC MASS

Composition:

Dense organic bone (Skull).



Physics:

g

High mass, structural collapse (catastrophic fall from canopy).

Outcome:

g

Gravity instantly pulls the heavy biological mass downward, initiating a dynamic slide down the unstable coal ridge.

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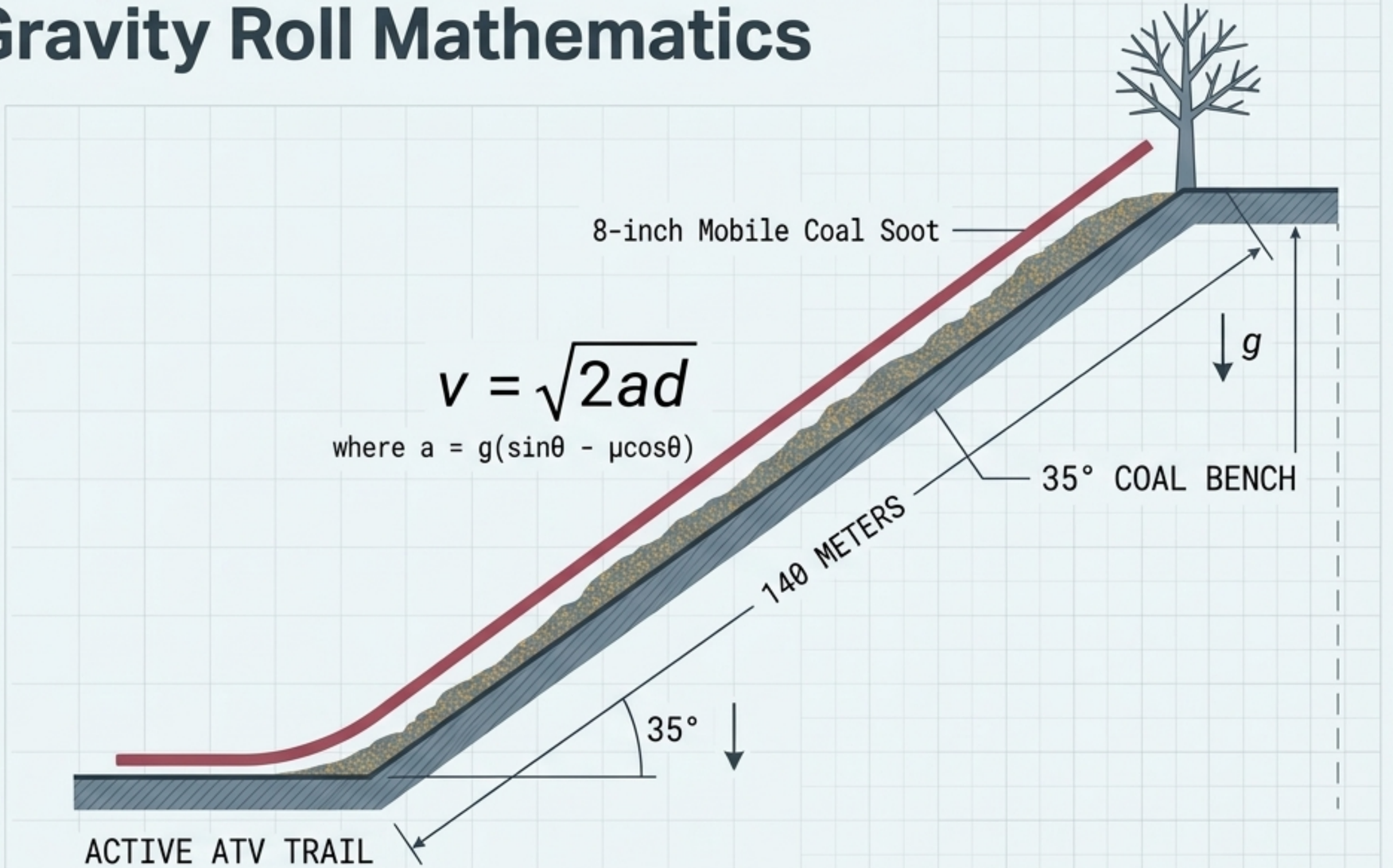
The 140-Meter Gravity Roll Mathematics

The Physics Breakdown

1. **The Matrix:**
   35-degree steep quarry face covered in 8 inches of loose, mobile coal soot.

2. **Taphonomic Reality:**
   Over 11 years of freeze/thaw cycles and downpours, heavy biological mass naturally breaks free from the tree catchment.

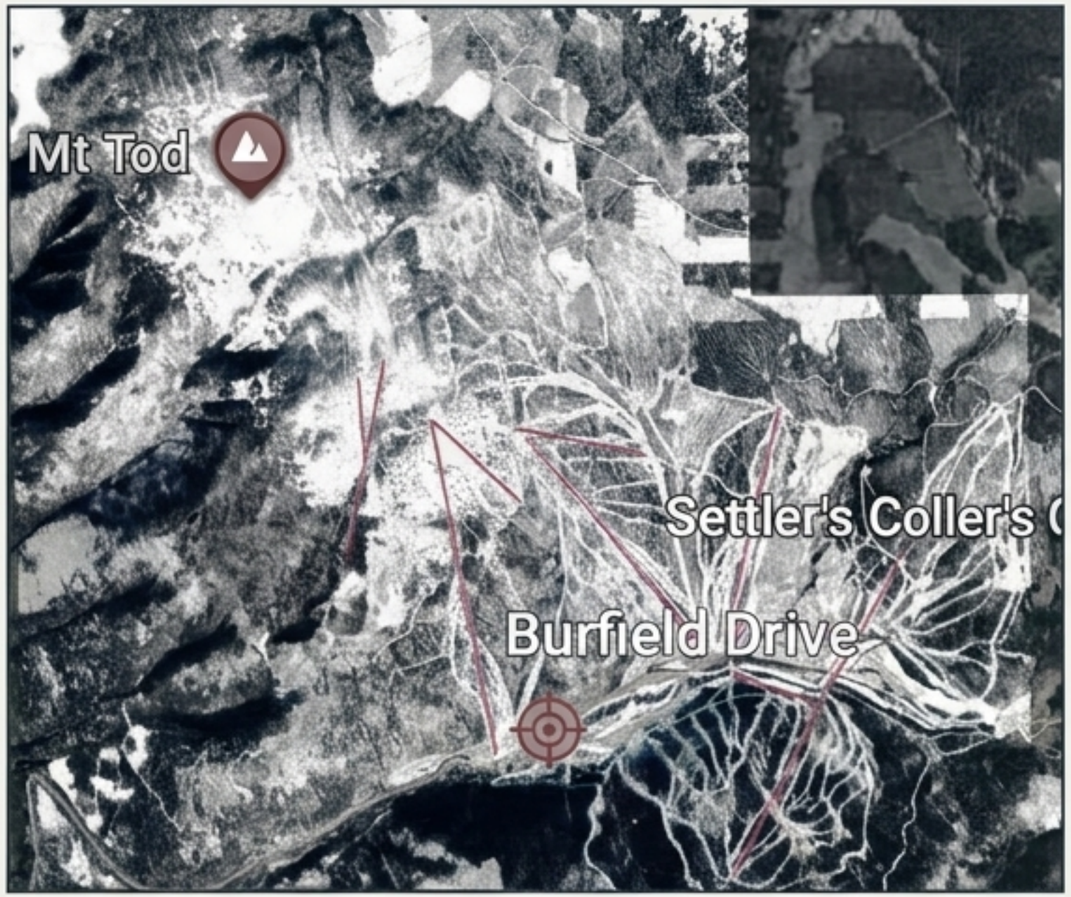
3. **Deterministic Conclusion:**
   It rolls exactly 140 meters downslope, clearing the soot bank, settling at the absolute baseline equilibrium point: the trail.



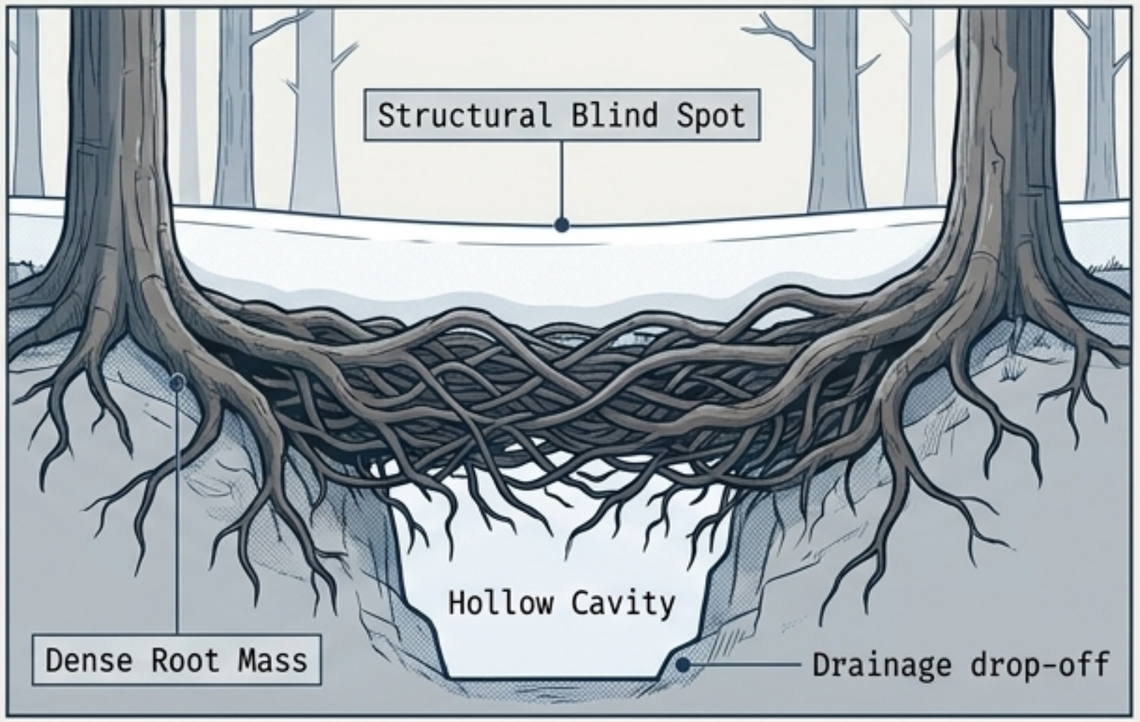
This deterministic trajectory mathematically disproves fears that remains were "planted" by foul play. The environment dictated the exact deposition.

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Environmental Masking: The Rootball Trap



Case Context: Ryan Shtuka (Sun Peaks, BC) & Liam Toman (Mont-Tremblant, QC). Both vanished in heavy snow (-26°C, active 8-inch snowfall).



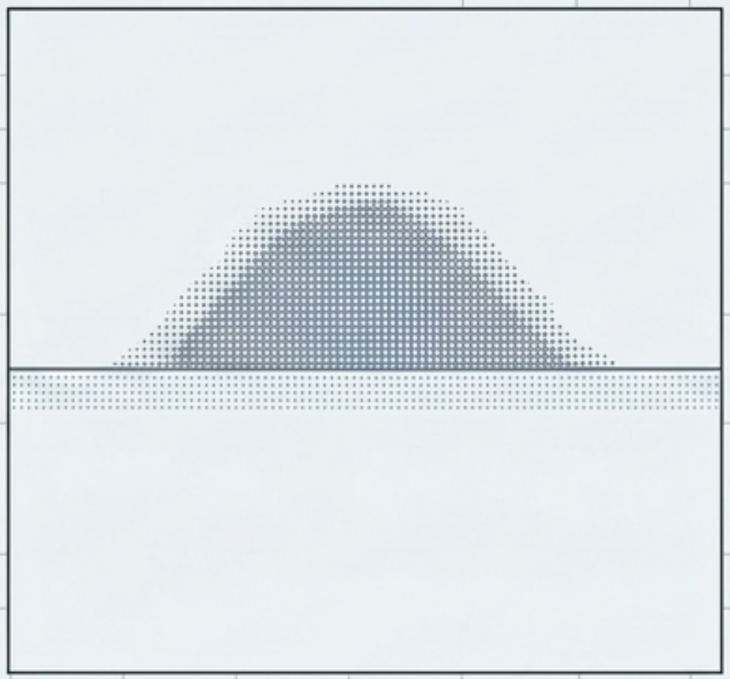
- 4-foot rootbound rootballs over drainage drop-offs create structural traps.
- Heavy snow creates a solid bridge over the cavity.
- Standard ground searchers walk directly over the void.
- Thermal (FLIR) cannot penetrate dense root mass and snow layers.

When remains are pulled beneath a root system’s mass, visual grids and canine assets yield a 0% Probability of Detection

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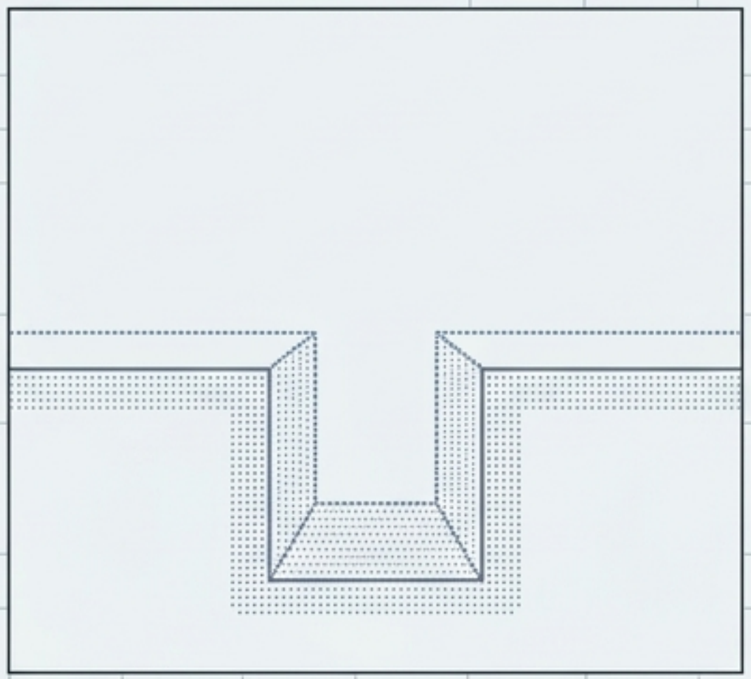
Digital Rootball Stripping (CSF Automation)

Standard LiDAR



Software misidentifies the thick, solid cluster of roots as actual terrain, creating a false floor.

Filtered LiDAR



The rootball is digitally erased, revealing the clandestine deep trench well beneath.

```
pdal Cloth Simulation Filter (CSF) execution:
> Lock CSF rigidness threshold = 3.0
> Point cleaning outlier multiplier = 1.5
```

By aggressively re-classifying high-slope vegetation anomalies, the HFMR engine computationally strips away 4-foot rootbound blind spots, exposing clandestine ditch-well geometries that physical searchers missed.

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Subterranean Physics: The Ebby Steppach Effect

The Forensic Complication:

A 12-inch structural conduit leading into a sealed sanitation vault alters decomposition venting entirely.

The Physics of Scent Deception:

1. Horizontal Venting

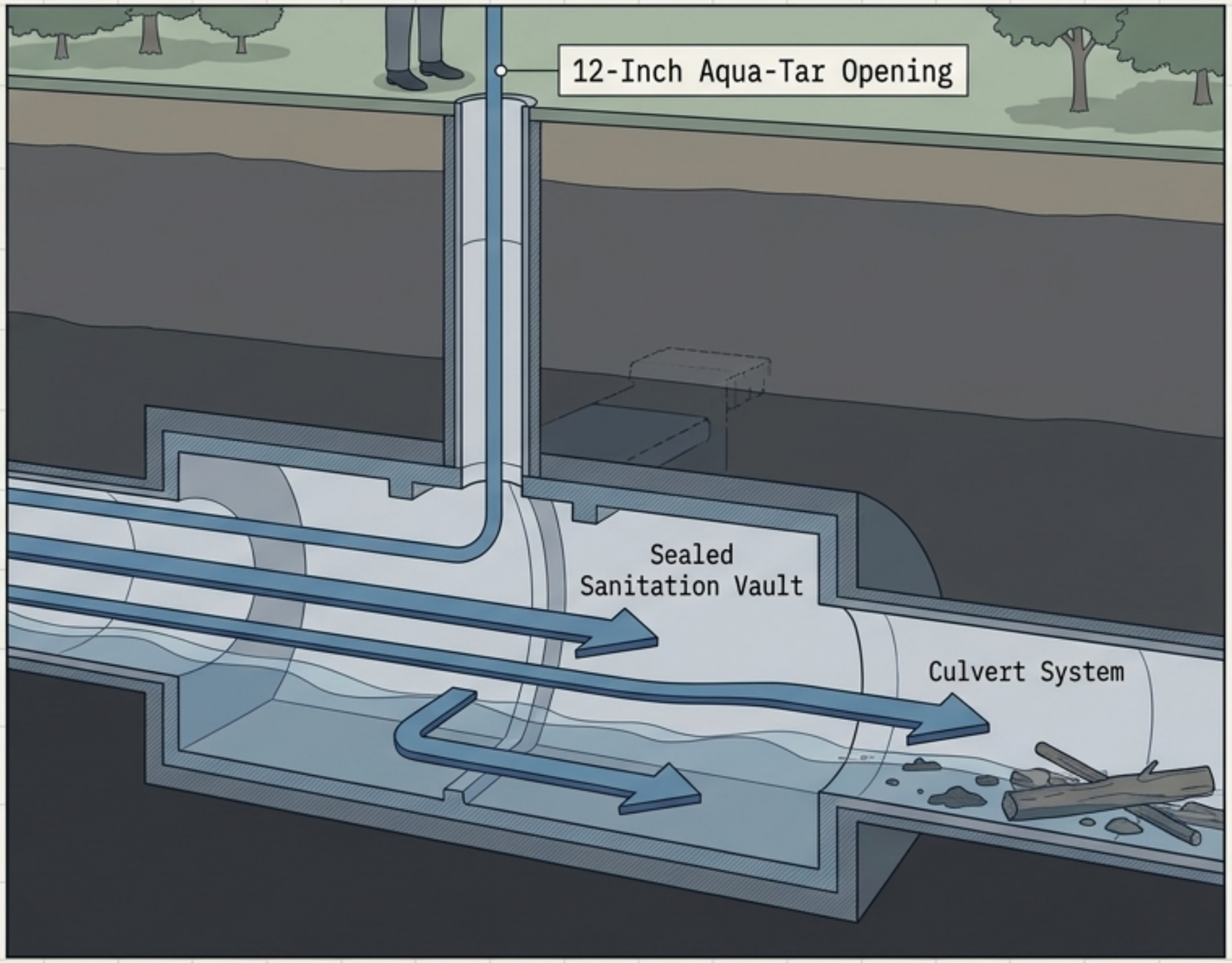
In a sealed vault, ambient airflow is drawn through the pipeline network. Gases do not rise vertically.

2. K9 Failure

Scent is carried horizontally downstream by subterranean drafts. HRD (Human Remains Detection) K9s walking directly over the surface opening show zero alerts.

3. Visual Failure

A 12-inch surface opening is too narrow for human searchers to peer down. Remains are trapped in logjams miles away.



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The Spatial Overlay Engine: Achieving 98%+ Confidence

Deterministic LiDAR

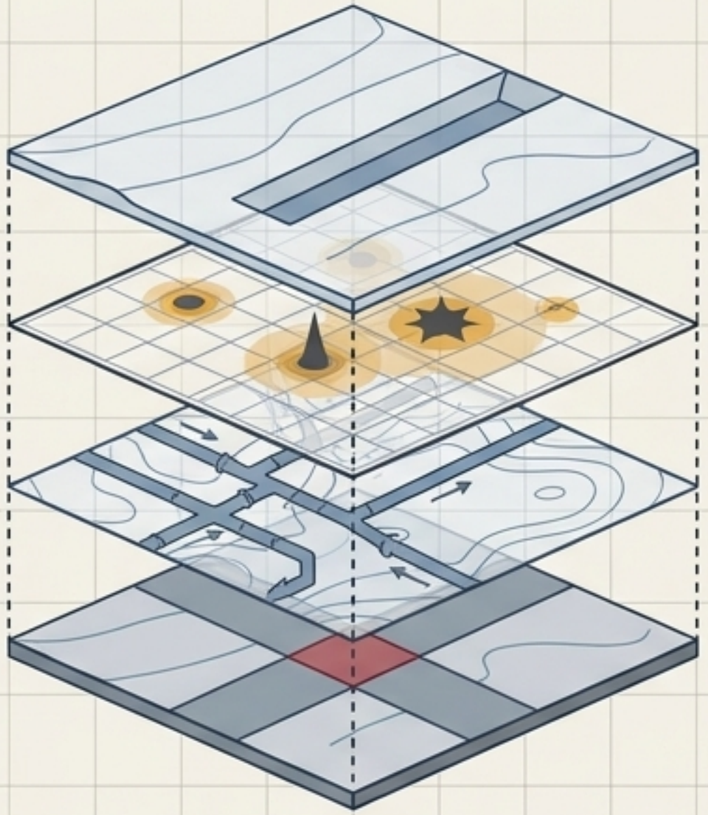
Bare-earth digital terrain models with rootballs stripped. (Identifies structural voids/trenches).

Predictive Metal Anomalies

Weighted grid matrices scoring metal detection spikes (phones, belt buckles, keys).

Hydrologic Pipeline Path

Mapping municipal sanitation pipes and drainage gradients to predict horizontal drift.



```
target_matrix = compute_forensic_probability(shape, mock_lidar, mock_metal, mock_pipes)
> High prob coordinates >= 95.0
```

We mathematically force the model to highlight exact cells where pipeline logic intersects physical obstructions and metal spikes. This is how we find the labyrinth.

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The Evidence Integrity Problem

The Problem



The Context:

Investigators are inundated with subjective tips, true-crime theories, and shifting narratives (e.g., “The Coyotes planted the remains”).

The Solution



The Requirement:

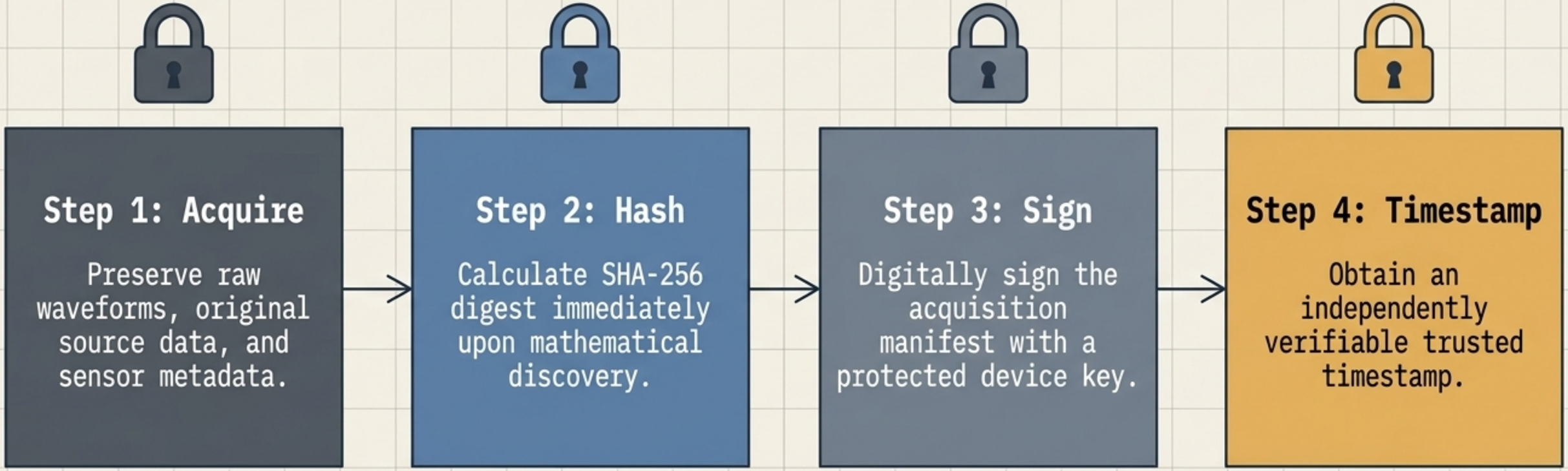
To be court-ready, and to secure reward priority ($5,000) before investigator contact, the digital timeline must be mathematically unassailable.

The Solution: Cryptographic Empirical Physics.

We must prove that a physical anomaly was mapped, and the specific coordinates identified, before the physical discovery on the trail.

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The Cryptographic Evidence Ledger



Hash Formula: $H_i = SHA256\ (H_{i-1} \,||\, D_i \,||\, M_i)$

Cryptography doesn't prove the object is brass; it proves the physics calculation pointing to the object has not been altered without detection.

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Locking the Immutable Witness

```
CASE_ID: FARBER-2015-PA
SUBJECT: Jesse Farber (Sharp Mountain Ridge)
TAPHONOMIC PATHWAY: 140-Meter Downslope Gravity Roll
GRAVITY PROFILE: v = sqrt(2ad) intersecting 8-inch soot catchment
TARGET GEOMETRY MATRIX: Subsurface Extraction Grid Crossing 98%+ Spatial Weights

=== CRYPTOGRAPHIC TARGET WITNESS LOCKED ===
Timestamp: 2026-10-08 14:00:00 UTC
SHA-256 Hash Signature: 9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08
```

Old Data. New Technology. New Leads.
Because old evidence deserves a second look.

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